

EDUCATION

Degree	Specialization	Institute	Year	CPI
B.Tech	Computer Science & Engineering	Amrita Bangalore	2025	8.01/10
12th CBSE	Physics, Chemistry, & Mathematics	Narayana College	2021	91.0%
10th ICSE	-	New Baldwins	2019	91.6%

WORK EXPERIENCE

- **Software Developer & DevOps Engineer** [Simnovus] Feb 2025 – Present
 - Developing and maintaining tools for large-scale **4G/5G telecom networking simulators**. Automating **simulator deployments, VM creation, and infrastructure management** using **Jenkins, Docker, Linux, and Shell scripting**. Working on **CI/CD pipelines, virtualization environments, system configuration, and simulator orchestration** while also contributing to internal tools and dashboards using **React, Angular, and GoLang**.
- **Machine Learning Intern** [Infosys Springboard] Oct 2024 – Dec 2024
 - Developed a **neural network-based multilingual translation system** supporting six languages using **Python, NLP techniques, and deep learning models**. Built an interactive **Streamlit web application** for real-time translation, implementing data preprocessing pipelines and pretrained language models while coordinating development tasks and presenting the final system.
- **Web Development Intern** [Zima Blue Management] May 2024 – Jul 2024
 - Led a **team of three developers** to design and deploy a responsive corporate website with optimized **UI/UX**. Developed frontend components using **HTML, CSS, and JavaScript** and deployed the application using **Firebase hosting**, ensuring scalable deployment, efficient performance, and maintainable architecture.

PROJECTS

- **Ternix – Privacy-First Desktop Terminal & Remote Session Manager** Mar–Jul 2026
 - Built a cross-platform **SSH & remote session manager** supporting **SSH, Telnet, Serial, local shells, RDP, and VNC** with multi-pane splitting (up to 6 panes) and tab tear-out into floating windows. Implemented an **encrypted SQLite vault (AES-256-GCM)**, **SFTP file manager** with visual **chmod editor**, and mobile session syncing using **Electron, React, TypeScript, xterm.js, and SQLite**.
- **Bloom – Radial Desktop Launcher & Local AI Workspace** Mar 2026
 - Designed a floating bud desktop launcher and focus workspace featuring **nested radial dial navigation**, local **Whisper AI voice dictation** (zero network/API calls), OS-native text-to-speech, **Eisenhower task matrix**, and depleting ring focus timer with procedural synth tones using **Electron, AI/Whisper, and JavaScript**.
- **DevFlow – Autonomous AI-Orchestrated Engineering Platform** Feb 2026
 - Developed an autonomous engineering platform connecting Jira and GitHub via **Model Context Protocol (MCP)**. Built an autonomous **FixAgent** powered by **Gemini CLI** to analyze tickets, search codebases, apply repairs, verify changes with shell commands, stream real-time logs/diffs (SSE), and open GitHub Pull Requests using **React, Node.js/Express, AI/Gemini, and MCP**.
- **Morphix – Cross-Platform File Conversion & Media Toolkit** Jan 2026
 - Created a desktop file conversion suite for documents, video, audio, images, and PDFs. Integrated universal engine selection, **PDF manipulation** (merge/split/compress/watermark via pdfjs-dist), **interactive image cropping/skia canvas**, **video trim/compression/subtitles** via fluent-ffmpeg, and **audio waveform trim** (wavesurfer.js) using **Electron, React, TypeScript, and FFmpeg**.
- **Real-Time System Resource Monitoring Dashboard** Jun-Jul 2025
 - Built a **system monitoring dashboard** to track CPU, memory, and network utilization across Linux systems. Developed backend monitoring scripts using **Python and Shell scripting** and visualized system metrics using **web-based dashboards**. Designed the system for **real-time monitoring and performance diagnostics**.
- **Road Safety and Accident Prevention using RF in Speed Limit Zones** Sep-Oct 2024

- Designed a **vehicle speed control and accident prevention system** using **Arduino Uno, RF transmitters/receivers, ultrasonic sensors, and embedded C**. The system automatically enforces **speed limits in restricted zones such as schools and hospitals** and detects obstacles to trigger emergency braking for enhanced road safety.
- Secure Multi Client-Server Communication using SSL and Hybrid RSA-AES** *Mar-Apr 2024*
 - Developed a **secure client-server communication framework** using **SSL/TLS with hybrid RSA-AES encryption**. Implemented **secure key exchange, digital signatures, and encrypted message transmission** to ensure confidentiality, integrity, and authentication. Built using **Python, socket programming, and cryptography libraries** to simulate secure communication between distributed network clients.
- Genetic Algorithm for Intelligent Vehicle Routing in Logistics** *Jun-Aug 2023*
 - Implemented a **genetic algorithm based optimization system** to solve the **Vehicle Routing Problem (VRP)** for logistics networks. Developed using **Python and evolutionary optimization techniques** to compute optimal delivery routes based on **distance, demand, travel time, and geographic coordinates**, reducing routing cost and improving delivery efficiency.

PUBLICATIONS

- Multi-Client Server Based Quantum Key Distribution** [*ICCCNT, IIT Indore*] *Jul 2025*
 - Presented at IIT Indore**, Designed a quantum-secure communication framework for distributed systems using **Quantum Key Distribution (QKD)** to enable scalable and secure key exchange between multiple clients and servers.
- Cloud-Based Real-Time Anomaly Detection in Network Traffic** [*ICOCT, Bengaluru*] *Jun 2025*
 - Implemented a **machine learning-based anomaly detection system** for network traffic monitoring using cloud infrastructure. Built real-time analytics pipelines for detecting suspicious activities in distributed network environments.
- CloudShare: Passwordless Cloud-Based File Storage & Sharing Framework** [*ICOCT, Bengaluru*] *Jun 2025*
 - Developed a **secure cloud storage framework** eliminating traditional password authentication using **asymmetric cryptography and challenge-response protocols**. Integrated secure file sharing with **AES encryption and controlled access mechanisms**.
- Development of a Robust Emergency Communication System Utilizing LoRa and GPS with Multi-Hop Routing** [*ICSSAS*] *Oct 2024*
 - Proposed a **long-range emergency communication system** using **LoRa modules and GPS-based location tracking**. Implemented a **multi-hop mesh routing architecture** for transmitting distress signals in disaster scenarios where conventional communication infrastructure fails.
- License Plate Detection and Recognition Using YOLOv8 and OCR** [*15th ICCCNT, IIT Mandi*] *Jun 2024*
 - Presented at IIT Mandi**, Developed a computer vision based **automatic license plate recognition system** using **YOLOv8 object detection** and evaluated multiple OCR engines including EasyOCR, PaddleOCR, and Tesseract for character extraction.
- Improvising Energy Efficiency of Heterogeneous Servers using AWS Services and Machine Learning Approaches** [*IEEE ICCCNT*] *Jun 2024*
 - Proposed a **machine learning-based approach** to optimize energy consumption in cloud infrastructure. Implemented regression models and deployed containerized workloads using **AWS EC2 and Docker** to improve resource allocation efficiency.

PATENT WORKS

- Pothole Detection and Repair System and Method Thereof** [*Indian Patent*] *2025*
 - Developed an **automated road maintenance system** capable of detecting potholes using image acquisition and processing techniques. Integrated **computer vision, geolocation tracking, and automated filler dispensing mechanisms** to identify pothole dimensions and perform real-time repairs.
- IoT-Enabled Pharmaceutical Inventory Management System for Hospital Pharmacies** [*Indian Patent*] *2024*
 - Designed an **IoT-based smart inventory system** to automate pharmaceutical stock management in hospital pharmacies. Implemented **barcode scanning, real-time inventory tracking, automated restocking alerts, and expiry monitoring** using **ESP32 microcontrollers, PHP, MySQL, and web-based dashboards** to reduce manual errors and improve operational efficiency.

CERTIFICATIONS AND ACTIVITIES

- **AWS Academy Certifications:** Cloud Architecting and Security Foundations 2024–2025
- **AI/ML Certifications:** Applied Generative AI & NLP, Machine Learning with Python, Artificial Neural Networks, Data Manipulation using NumPy & Pandas [Udemy] 2024–2025
- **Cybersecurity:** Recon for Bug Bounty, Penetration Testing and Ethical Hacking [Udemy] Aug 2024
- **Workshops:** DevOps Workshop (CI/CD, Containerization), Computer Vision Research Workshop 2023
- Participated in **Live in Labs**, solving real-world socio-economic problems in a rural village Jan 2024
- **Office Bearer — EPIC Club** (Videography Club), Amrita Vishwa Vidyapeetham 2023–2024
- **Web Development Certification** [Acmegrade] Apr 2022
- Volunteered in **AYUDH** social service initiative at Amrita Hospital, Faridabad Aug 2022
- **Black Belt in Karate — Shitō-ryū Style** 2020

SKILLS

- **Programming:** Python, Go, Java, C/C++, JavaScript, SQL, Shell Scripting
- **Cloud & DevOps:** AWS, Docker, Jenkins, CI/CD Pipelines, Linux, Virtualization
- **Web Development:** React, Angular, HTML, CSS, Flask, Streamlit
- **Systems & Networking:** TCP/IP, SSL/TLS, Cryptography, Distributed Systems, Network Simulation
- **IoT & Tools:** ESP32, Arduino, Wireless Communication, Git, Jupyter, VS Code, LaTeX, MySQL
- **Languages:** English, Telugu, Kannada, Hindi, Tamil, German (Beginner)